

Lab-3 Ex4 - Your Classmate Needs Your Help!

Your classmate, Peter, wants to create a simple C program that obtains *an integer* from the user. It should determine whether the input integer is:

- zero; or
- a positive odd number; or
- a positive even number; or
- a negative odd number; or
- a negative even number.

and then print the result to the screen. However, he feel miserable because his program doesn't even compile! Here's his program:

```
1  #include <stdio.h>
2
3  int main(void) {
4      int x;
5      scanf("%d", x);
6
7      if (x = 0)
8          printf("Zero");
9      else (x % 2 = 0)
10         printf("Positive even number");
11     else
12         printf("Positive odd number");
13
14     return 0;
15 }
```

Can you copy his program and help him **DEBUG** the program? Make sure it compiles and the output is correct. Assume all inputs are valid and in the range of -1000 to 1000.

Hint:

Recall that all even numbers, if divided by 2, will have a remainder of 0. So in C, for any even number x , $x \% 2 == 0$

User-input numbers are ***bold and italic*** in the following sample runs.

Sample run #1:

-99

Negative odd number

Sample run #2:

0

Zero

Sample run #3:

5

Positive odd number

Sample run #4:

-20

Negative even number